

Unit 6, Lesson 4: Converting Mass and Area

Benchmark

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Learning Target

- ☐ I can solve problems involving conversion of mass and area measurements across different measurement systems.

6.4.1 Warm-Up: Orca vs. Human

Echolocation is the use of sound waves and echoes to navigate and find food, as seen in bats and whales.

An orca can "see" objects via echolocation from 150 meters (m) away.



A human with 20/20 vision can clearly see something from 20 feet (ft) away.

1. How many times farther can an orca "see" compared to a human?

6.4.2 Guided Instruction: Conversions of Mass and Weight

In the metric system of measurement, the most common units for measuring mass include milligrams (mg), grams (g), and kilograms (kg). In the customary system, the most common units for measurement for weight include ounces (oz), pounds (lb), and tons (t).



Weight vs. Mass

Weight and mass are often used interchangeably, but they are not quite the same, as you'll learn in science. The mass of an object never changes, but its weight changes depending on gravity. For example, your weight on Earth and your weight on the moon would be different, while your mass would remain the same. Since you're in math class here on Earth, mass and weight will be used interchangeably.

1. With your partner, discuss some real-world examples of measuring mass.

Several common conversion factors for weight and mass are shown.

1 ounce (oz) = 28.35 grams (g)
1 pound (lb) = 454 grams (g)
1 kilogram (kg) = 2.204 pounds (lb)

Each conversion factor can be expressed as a ratio or unit rate. For example, $\frac{454 \text{ grams}}{1 \text{ pound}}$ is the unit rate of grams to pounds.

2. Dr. Corlett, a veterinarian, uses weight in kilograms to determine safe dosages of medicine to prescribe for animals.

One of her patients, Opie, is a golden retriever that weighs 85 pounds (lb).

Complete the calculation below to convert Opie's weight to kilograms (kg). Round to the nearest tenth.

$$\frac{85 \text{ lb.}}{1} \times \frac{\boxed{}}{\boxed{}} = \boxed{} \text{ kg}$$

3. Why is it important to include the units in conversion calculations?
4. The table includes information about several of Dr. Corlett's patients.

Name	Breed	Weight (lb)
Kilo	French Bulldog	16
Minnie	Maltese	4 $\frac{1}{2}$
Cesar	Irish Terrier	23
Princess	Pug	14.2

A certain medication Dr. Corlett uses cannot be prescribed to dogs that weigh less than 10 kg. For which dogs does Dr. Corlett need to be careful not to prescribe this medication?

6.4.3 Collaboration: Conversions Within and Between Systems

Conversion factors that can be used to convert mass between and within measurement systems are shown.

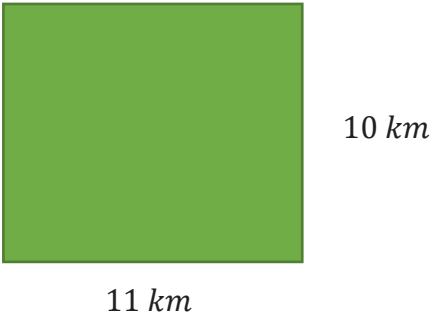
Converting from Customary to Metric	Within the Customary System
1 ounce = 28.35 grams	1 pound = 16 ounces
Converting from Metric to Customary	Within the Metric System
1 kilogram = 2.204 pounds	1 kilogram = 1000 grams

1. The weights and masses of four objects are given in the table. Work with your partner to complete the table by using the conversion factors listed. Round to the nearest thousandth when necessary.

Object	Ounces	Pounds	Grams	Kilograms
Bag of potatoes		5		
Pencil	0.3			
iPhone			194	
Bicycle				9

6.4.4 Guided Instruction: Conversions of Area

1. What does area measure?
2. What type of units are used to measure area?
3. Luciana is a wildlife biologist who studies the habitat of the Florida Panther. Florida Panthers are tracked using radio collars to determine the area in which the panther lives. The rectangular habitat of one of the panthers is shown, where the dimensions are given in kilometers (km).



Two students used different methods and conversion factors to find the area of the panther’s habitat in square miles (mi²).

Keyshaun’s Work	Harper’s Work
1 mi = 1.61 km $\frac{11\text{ km}}{1} \times \frac{1\text{ mi}}{1.61\text{ km}} \approx 6.832\text{ mi}$ $\frac{10\text{ km}}{1} \times \frac{1\text{ mi}}{1.61\text{ km}} \approx 6.211\text{ mi}$ $6.832\text{ mi} \times 6.211\text{ mi} \approx 42.43\text{ mi}^2$	$11\text{ km} \times 10\text{ km} = 110\text{ km}^2$ $1\text{ mi}^2 \approx 2.5921\text{ km}^2$ $\frac{110\text{ km}^2}{1} \times \frac{1\text{ mi}^2}{2.5921\text{ km}^2} \approx 42.44\text{ mi}^2$

- a. Draw an asterisk (*) next to the conversion factor that each student used.

- b. Discuss each student’s work with your partner. Then, write a brief description of how each person found the area of the panther’s habitat in square miles.

Keyshaun’s Work	Harper’s Work

- c. Keyshaun and Harper used different conversion factors, but still found the same answer. How are their conversion factors related?
- d. Why did Keyshaun and Harper use the symbol $\approx$ in their calculations?

6.4.5 Try It: Conversions of Area

Acres are another common unit used to measure large areas of land. An acre is almost the size of a football field, or the size of about 16 tennis courts combined.

- 1. The Nature Conservancy’s Disney Wilderness Preserve covers approximately 11,500 acres of land. Convert the area of the preserve to square miles and to square kilometers. Round to the nearest hundredth when necessary.

1000 acres = 1.5625 square miles
100 acres = 0.405 square kilometers

Acres	Square Miles	Square Kilometers
11,500		

6.4.6 Your Turn!

1. Makayla is a real estate agent. A client is interested in an apartment with a primary bedroom that is rectangular in shape with dimensions of 15 feet (ft) by 11 ft. The client lives in Brazil and has asked for the area of the primary bedroom to be given in square meters (m^2). Makayla found the following conversions:

$$1 \text{ ft} = 0.305 \text{ meters (m)}$$

$$10.7584 \text{ square feet (ft}^2\text{)} = 1 \text{ square meter (m}^2\text{)}$$

What is the area of the primary bedroom in square meters? Round to the nearest thousandth.

2. Leah lives in Wales. In Wales, body weight is measured using an old customary unit called stone. The stone has been a measurement of mass since before the Romans. Leah is packing to go on vacation and is allowed 25 kilograms (kg) of luggage on the flight. Leah uses her bathroom scale to measure the weight of her luggage. The luggage weighs 3 stones and 3 pounds. Determine if Leah's luggage is over the weight limit.

$$1 \text{ stone} = 14 \text{ pounds}$$

$$1 \text{ kilogram} = 2.204 \text{ pounds}$$

6.4.7 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate how interesting you find the lesson, along with how much sense the lesson made to you, for today's learning target.

I can solve problems involving conversion of mass and area measurements across different measurement systems.

